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LESSON 7

Income-Based Modeling

Learning Objectives

In this lesson you will be required to:

1. Develop a simple additive model for determining capitalization rates to assist in the valuation of apartment buildings
2. Test the model, using the sales provided, to check the accuracy of the capitalization rate estimated and its usefulness in estimating sale price
3. Test the model by comparing its performance to traditional single property methods of estimating capitalization rates

ASSIGNED READING

1. UBC Real Estate Division. 2020. *BUSI 444 Course Workbook*. Vancouver: UBC Real Estate Division.
Lesson 7: Income-Based Modeling
2. UBC Real Estate Division. 2019. *Advanced Computer Assisted Mass Appraisal*. Vancouver: UBC Real Estate Division.
Chapter 10: Specifying and Calibrating the Income Approach

RECOMMENDED READING

1. *Winnipeg.ca* Property Assessment. Once there, search for guides on their approach to the valuation of non-residential properties.



Selected readings can be found under Online Readings on your Course Resources webpage. Please note that recommended readings are not tested on the exam, but are provided for further research into these topics.

INSTRUCTOR'S COMMENTS

The methodology used in this lesson will be the same as that used to develop models in the previous lessons:

1. Describe an appropriate general model to use and state this model using standard mathematical symbols.
2. Review the variables in the database and identify those which are suitable for this type of model.
3. Examine the potential independent variables for relationships with each other and with the dependent variable using graphic analysis, cross tabulation tables, and correlation analysis.
4. (a) Create any transformations necessary to make variables suitable for the chosen model structure.
(b) Create any additional transformations required to remove problems of collinearity identified in Step 3.
5. Repeat Step 3 with the new transformed variables.
6. List a final group of potential variables for calibration.
7. Calibrate the model using an appropriate method.
8. Test and evaluate the model for use.
9. State your conclusions as to model quality.

Before beginning this lesson, a review of some of the concepts from Chapter 10 of Advanced Computer Assisted Mass Appraisal will be useful. The text illustrates a spreadsheet analysis of gross income and a tabular analysis of overall capitalization rates for four groups of properties. These methods are typical of those used by single property appraisers or assessment staff in small jurisdictions where the amount of data available for commercial and multi-family residential properties is limited. However, multiple regression analysis techniques can be used by assessment staff in both large and small jurisdictions to estimate units of comparison for both the market comparison and income approaches to value for these property types. In fact, these techniques will usually result in more accurate estimates of the capitalization rate, GIM, expense ratios, and gross income than the methods used by single property appraisers. Multiple regression analysis allows an appraiser to make full use of all data available rather than simply stratifying the properties and computing means or medians of the units of comparison. Since assessment staff tend to have more data available and more staff time to devote to data collection and analysis, it is logical that this data should be put to its best use in creating better valuation models for income producing properties.

The database entitled Apartment contains property information on 30 apartment building sales from one municipality. You should have already copied the database file (Apartment.sav) to your computer. Open the Apartment database.

The variables on the database are shown below:

Variable Name	Description	
saledate	date of sale (yyyymmdd)	1
price	sale price	2
nbhd	nhbd # (1-4)	3
units	total number of units	4
bach	number of bachelor units	5
rent_b	mean rent per bachelor unit	6
onebed	number of 1 bedroom units	7
rent_one	mean rent per 1 bedroom unit	8
twobed	number of 2 bedroom units	9
rent_two	mean rent per 2 bedroom unit	10
threebed	number of 3 bedroom units	11
rent_thr	mean rent per 3 bedroom unit	12
gross_in	potential gross income (annual)	13
vacancy	vacancy allowance (2%)	14
eff_inc	annual effective income (gross - vacancy)	15
expenses	mean annual expenses	16
area	building area in square feet	17
perimetr	building perimeter feet	18
story	number of stories	19
yrbuilt	year built	20
effage	effective age	21
conditn	condition (1=poor, 2=average, 3=good)	22
flrhght	mean floor height	23
deprec	depreciation	24
noi	effective income - expenses	25
heat_own	heat paid by owner (1=yes)	26
heattype	heating:1=hot water, 2=electric	27
und_park	underground parking (1=yes)	28
elevator	existence of elevator (1=yes)	29
lotsize	lot size in square feet	30

SPECIFYING THE MODEL

In this lesson you will be using a database of apartment building sales. You will develop and test a model to estimate the overall capitalization rate (often simply referred to as the “cap rate”) for all of the apartment buildings in the community. The model will use the income approach to valuation. The general model is:

$$MV = NOI \div OAR$$

where:

MV = estimated market value;

NOI = net operating income; and

OAR= overall capitalization rate

This model assumes that both the net operating income and the overall capitalization rate are available for all properties that will be valued with the model. It should be noted that this information is rarely available for all properties, and in cases where it is unavailable, the capitalization rate and net operating income must be estimated from the sold properties.

Assessment staff will generally have the gross income for most properties. Models can be constructed to estimate vacancy loss, expense ratios, and gross income using the data collected from the owners who complete the assessor's income and expense returns. This will allow an appraiser to develop an estimate of net operating income for those properties where it is not reported. In addition, the capitalization rate can be estimated using the available data and the sales information. The remainder of this lesson will demonstrate the development of an additive multiple regression model to estimate the capitalization rate and compare the result to that obtained using the traditional single property appraisal technique.

Equation 10.7 in Chapter 10 of Advanced Computer Assisted Mass Appraisal shows an additive model form for estimating the capitalization rate. The theoretical basis for this model specification is included there. The properties in this lesson are all apartments and therefore additional variables will be required (e.g., number of units, suite size, amenities such as elevators and underground parking). In addition, this database does not contain some information used in the model in the manual (e.g., restaurants, convenience stores), but does include some additional data that may be useful (e.g., location, land to building ratio).

When developing the specifications for the model, we must rely on appraisal and economic theory. The main components that make up a discount or capitalization rate are:

- risk free rate;
- allowance for risk;
- allowance for illiquidity; and
- allowance for investment management

In addition, the specific factors that affect real estate investment are given as follows:

- the discount rate (which varies with risk and remaining economic life);
- expected rate of change in net income;
- the percentage of income attributable to land (land requires no depreciation);
- location;
- condition;
- size; and
- land/building ratios.

Using this information, and the list of variables available in this database, the following general model can be specified for the overall cap rate (where b_0 is the constant and the b_x terms are the coefficients of the variables in the model):

$$\text{OAR} = b_0 + (b_1 \times \text{LOCATION}) + (b_2 \times \text{AGE}) + (b_3 \times \text{UNITS}) + (b_4 \times \text{SUITE SIZE}) + (b_5 \times \text{LANDRATIO}) + (b_6 \times \text{AMENITIES})$$

REVIEW AND EXAMINE THE VARIABLES

The next step in the model development process is to review the variables available in the database and group them according to the factor in the general model that they represent. Some characteristics in the model may be represented by more than one variable and sometimes it is necessary to use more than one variable to correctly represent the characteristic or factor needed in the model.

FACTOR	VARIABLE(S)
Location	nbhd
Age or Condition	yrbuilt, effage, conditn, deprec
Risk	vacancy, expenses, units
Management	units
Size	area, units
Land to Building Ratio	lotsize, area
Amenities	elevator, und_park, story

Before you begin to make changes to the database, you should familiarize yourself with the property characteristics contained in this database. Use Analyze → Reports → Case Summaries to produce a statistical summary with the following variables:

saledate
 price
 units
 gross_in
 expenses
 yrbuilt

The results indicate that the sale date ranges from January 31, 2015 to February 27, 2017 – a span of 25 months. The number of units per apartment building ranges from 4 to 120 and, as a result of these size differences, price, gross income, and expenses also demonstrate a great deal of variation. Therefore, it will be necessary to develop some standard measures to compare data.

Apartment buildings range in age from 25 to 50 years (a current year of 2018) with a median age of about 32 years.

Use Analyze → Descriptive Statistics → Frequencies to examine the following variables. Click Graphs → Legacy Dialogs → Histograms to display the histograms.

rent_b
 rent_one
 rent_two
 rent_thr

The rents charged within each category of bachelor, one bedroom, two bedroom, and three bedroom units should be comparable if the buildings are of comparable quality. If there are any significant differences, the characteristics of the particular property should be examined more closely by accessing the property in the Data Editor window. You should find the rents to be appropriate in this database.

Use Analyze → Descriptive Statistics → Frequencies to examine the following variables, again viewing their histograms as well:

units
 area
 yrbuilt
 effage
 lotsize

The range of the number of units is large (4 to 120) but more than 50% of the observations have fewer than 20 units.

Examination of the building area variable, area, shows results consistent with a database concentrated in smaller buildings. Most of the buildings have small building areas, and more than 50% of them have less than 15,400 square feet. Except for one very large building, the areas range from 3,460 to 60,531 square feet.

The buildings are of comparable ages with roughly 80% built between 1977 and 1992, a range of 15 years. A significant number of buildings (5) were built in 1991 alone. Thus, it can be concluded that this is an older, more established community with few new apartment buildings being constructed.

The median effective age is 31½ years and, as indicated in the histogram, the distribution of effective age is relatively even.

Except for one very large lot, the lot sizes are comparable, ranging from 5,712 square feet to approximately 55,000 square feet. About 60% of the sales have lot sizes of less than 27,000 square feet.

TRANSFORMATIONS

Appraisers who work with this type of property generally classify them into two types: one for buildings with a small number of units, which are normally managed by the owner and purchased as a long-term investment; the other for buildings with a large number of units which are normally managed by a professional rental agency, and held for a much shorter investment period. When developing units of comparison, a single property appraiser would divide the database into two groups to represent these two different market motivations. Since the database in this lesson is relatively small, for the purposes of mass appraisal, it will be retained as a single database.

There are two ways to determine if the number of units is significant in developing the capitalization rate. One would be to simply include the number of units as a variable. The other would be to create a binary variable to split the number of units into two groups. Since small investors usually prefer a building with fewer than 20 units, and as the median number of units is 18.5, we will create a binary variable that will have the value one if the number of units is 18 or less and zero if the number of units is greater than 18. This binary variable will create two groups of equal size.

To account for variations in scale among apartment buildings, additional transformations will be useful. For example, sale price per unit or sale price per square foot will allow comparisons to be made between large and small buildings. This is also true of the income variables `gross_in` and `noi`. The variable `eff_inc` is not useful, because it was created using a common vacancy ratio for all sales; `eff_inc` is simply 98% of the gross income (`gross_in`).

The direct capitalization rate and gross income multiplier are useful measures of the return investors anticipate in determining the sale prices they are willing to pay. The capitalization rate is also required; it will be the dependent variable in our model.

The ratio of lot size to building area and average size of units will illustrate whether there are any sales which are influenced by factors other than income. For example, a large ratio of lot to building size may suggest the potential for redevelopment; this potential could be reflected in the sale price.

Expense ratios should also be examined for any outliers.

The following transformations should be run to create the return measures discussed above:

COMPUTE <code>sp_unit</code> = price / units.	(sale price per unit)
COMPUTE <code>sp_sqft</code> = price / area.	(sale price per square foot)
COMPUTE <code>cap_rate</code> = noi / price.	(capitalization rate)
COMPUTE <code>gim</code> = price / <code>eff_inc</code> .	(gross income multiplier)
COMPUTE <code>lot_area</code> = lotsize / area.	(ratio of lot size to building area)
COMPUTE <code>area_unit</code> = area / units.	(building area per unit)
COMPUTE <code>exp_egi</code> = expenses / <code>eff_inc</code> .	(expense ratio)
COMPUTE <code>gi_unit</code> = <code>gross_in</code> / units.	(gross income per unit)
COMPUTE <code>noi_unit</code> = noi / units.	(net operating income per unit)

Open a new syntax file, name it `Apartment.sps`, and create the above transformations. Before continuing, you should check the results of these transformations. You should use Analyze → Reports → Case Summaries to examine the transformations. For example, for `sp_sqft`, enter price, area, and `sp_sqft` as the variables and manually calculate some of the values to check the results (Note: given the relatively small database, you could also list the results using Analyze → Descriptive Statistics → Frequencies and select various variables to display).

The variables `nbhd` and `story` are nominal variables and `conditn` is an ordinal variable. These should be transformed into binary variables for use in multiple regression analysis. Return to the syntax file and add the following transformations:

```
COMPUTE nbhd1=nbhd.
RECODE nbhd1(1=1)(else=0).
COMPUTE nbhd3=nbhd.
```



```

RECODE nbhd3(3=1)(else=0).
COMPUTE nbhd4=nbhd.
RECODE nbhd4(4=1)(else=0).
COMPUTE conditn1=conditn.
RECODE conditn1(1=1)(else=0).
COMPUTE conditn3=conditn.
RECODE conditn3(3=1)(else=0).
COMPUTE story2=story.
RECODE story2(2=1)(else=0).
COMPUTE story4=story.
RECODE story4(4=1)(else=0).
COMPUTE sml_unit=units.
RECODE sml_unit(lo thru 18=1)(else=0).

```

As can be seen from these transformations, the reference values will be 2 for nbhd, 2 for conditn, and 3 for story. These values are the most common for their respective variables.

Note that the final transformation creates the variable sml_unit which splits the apartments into two groups as discussed earlier (1 for apartments with 18 or fewer units and 0 for apartments with more than 18 units).

The following information was obtained from the Analyze → Descriptive Statistics → Frequencies window, by clicking Statistics and selecting the desired statistics:

Statistics					
	Median	Minimum	Maximum	Percentiles	
				25	75
sp_unit	61,436	42,300	90,721	53,750	72,589
sp_sqft	78.83	42.76	113.72	63.22	97.27
cap_rate	.0550	.0382	.0842	.0488	.0665
gim	11.20	7.16	15.73	9.12	12.78
lot_area	1.081	.800	1.959	.930	1.263
area_unit	797.96	625.33	1,167.88	731.24	866.50
exp_egi	.3905	.3608	.4321	.3740	.4001
gi_unit	5,911.43	4,800.00	7,407.60	5,525.14	6,411.05
noi_unit	3,551.16	2,671.40	4,619.20	3,273.79	3,835.20

The range of values in the table above is reasonable considering the large range in units. There are no outliers.

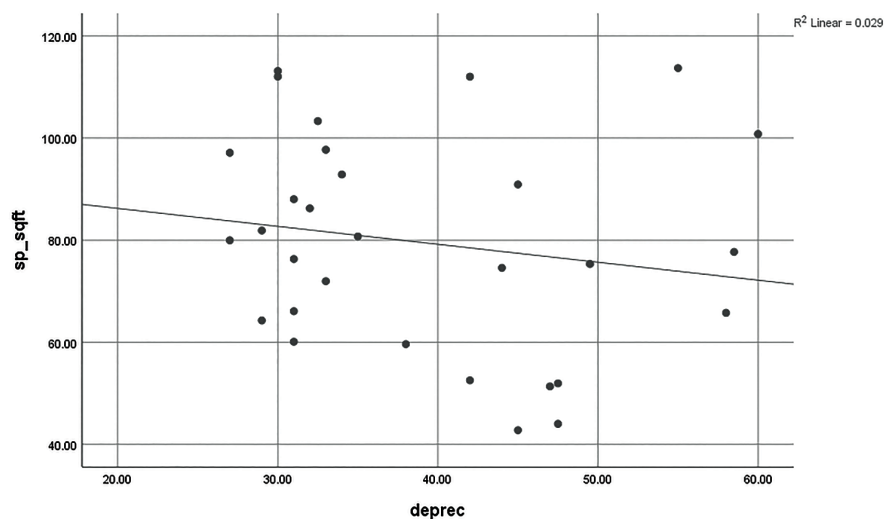
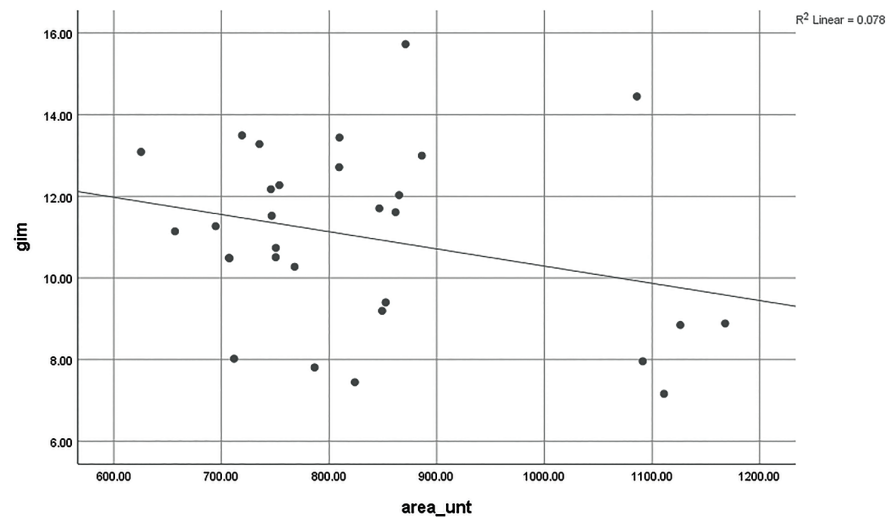
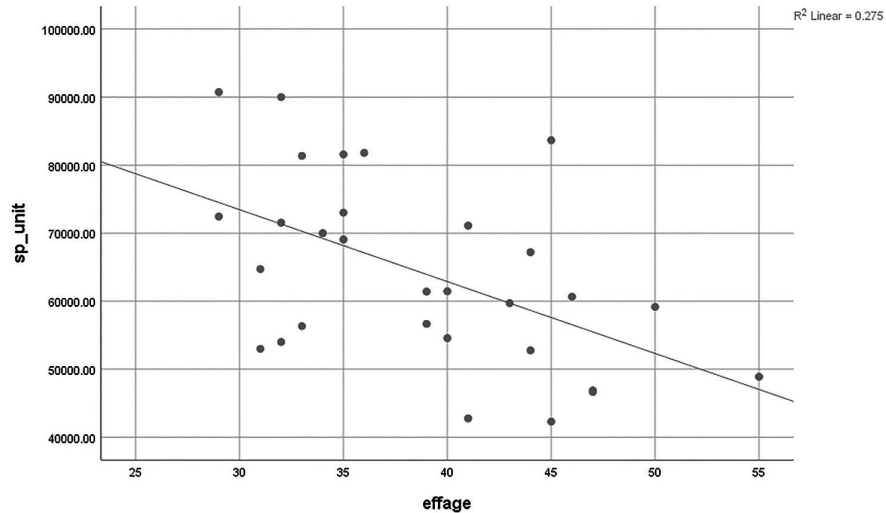
MULTIVARIATE ANALYSIS – RELATIONSHIPS BETWEEN VARIABLES

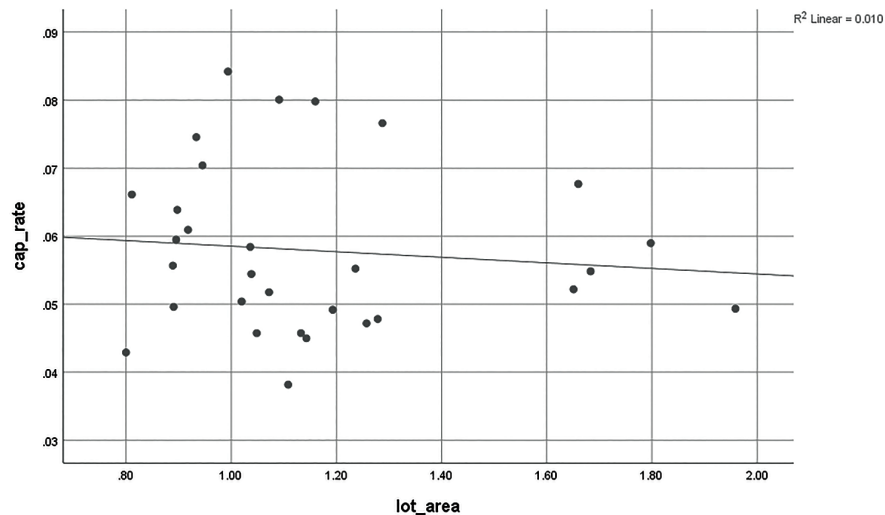
In this section we will examine the relationships of the variables to the sale price and other measures of return. The statistical tools for measuring relationships between variables include scatterplots, correlation coefficients, boxplots, crosstabulation tables, and simple linear regression.

Scatterplots

Scatterplots provide an efficient method of examining relationships among quantitative variables. Run scatterplots with `sp_unit`, `gim`, `sp_sqft`, and `cap_rate` as the Y (dependent) variable. The X (independent) variables should include `effage`, `area_unit`, `deprec`, and `lot_area`. Fit a linear regression line and display the R-squared statistic.

Four sample scatterplots are shown below:





Recall that the relationship between two quantitative variables can be expressed as a Pearson correlation coefficient (r) or its square (R^2). IBM SPSS scatterplots show the latter; however, taking the square root will produce the corresponding correlation coefficients. Alternatively, one can run a correlation analysis (as in previous lessons) to obtain the correlation coefficients. In any case, below is a summary of the correlation coefficients between some key variables:

Correlations

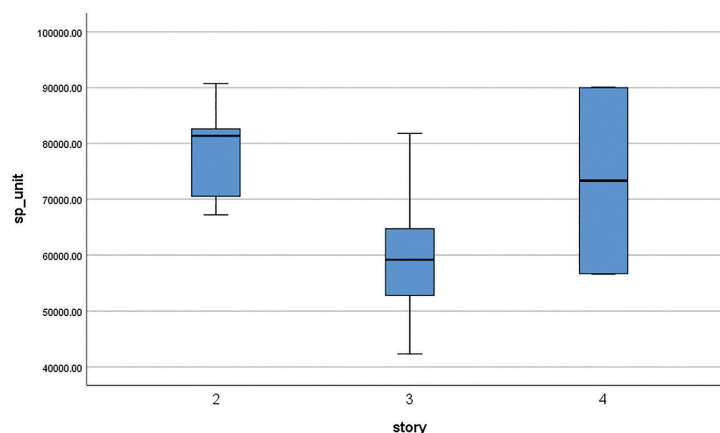
	effage	deprec	lot_area	area_unit
sp_unit	-.525	-.040	.085	-.091
sp_sqft	-.600	-.169	.153	-.619
cap_rate	.744	.109	-.101	.377
gim	-.692	-.031	.098	-.280

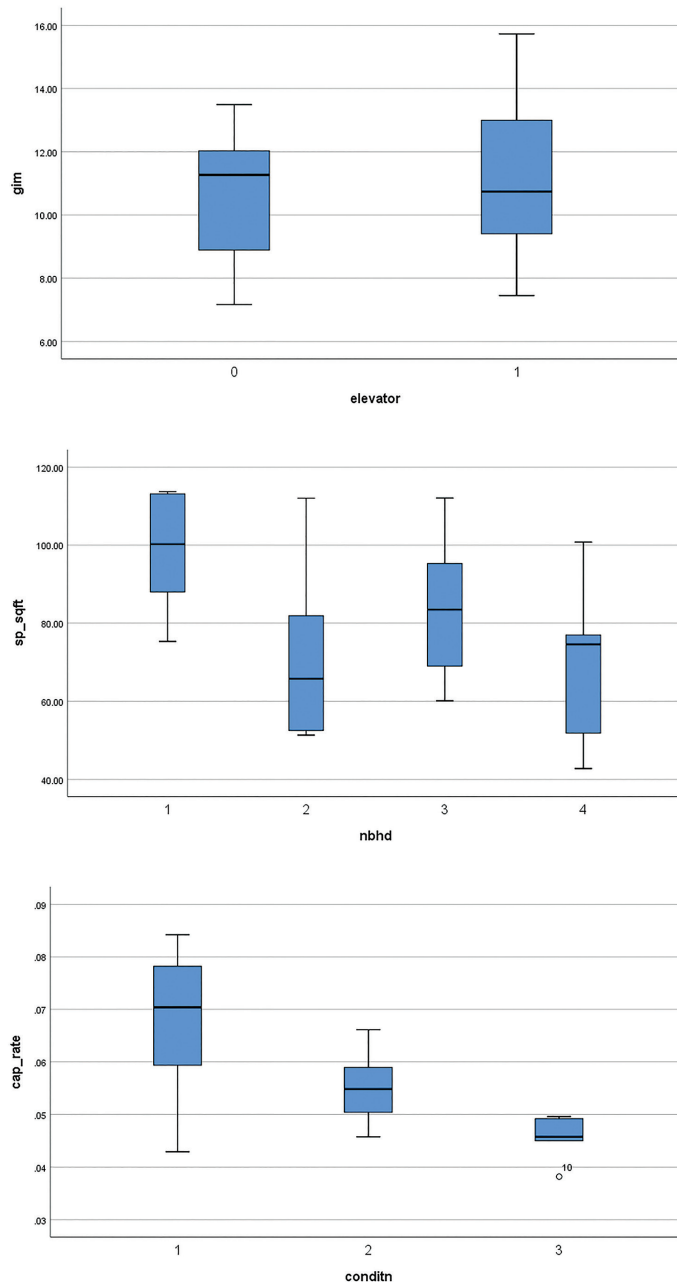
The plot slopes and correlation coefficients indicate that lower prices per unit or square foot and higher capitalization rates (leading to lower prices) are associated with older buildings. Relationships are strongest between effage and the dependent variables. There appears to be no relationship between the return measures and the ratio of land to building area.

Boxplots

Boxplots can be used to examine the relationship between the return measures (continuous variables) and discrete independent variables.

Examine the same dependent variables as in the scatterplots above (sp_unit, sp_sqft, cap_rate, and gim) with nbhd, story, conditn, und_park, and elevator.





All of the measures of comparison show a distinct variation as the values of the discrete variables change. This is true for all of the discrete variables except elevator, which only shows minor variations in the measures of comparison for properties with elevators compared to those without this feature.

As expected, because condition is an ordinal variable there is a progressive relationship between the values of condition and the units of comparison. That is, as condition improves from 1 (poor condition) to 3 (good condition), the median unit of comparison moves in the expected way, that is, cap_rate decreases (similarly, if sp_unit is the measure of comparison, as condition improves, sp_unit increases). Also note the greater variability in the return measures where the condition is poor; this likely indicates a greater degree of risk associated with renting properties in poor condition.

Correlation Analysis

In the previous two sections we have seen how some of the variables are related to cap_rate. Because some of the relationships are nonlinear, such as nbhd and story, we will not completely

rely on correlation analysis in the variable selection process. We will primarily use this technique to identify which of these variables are correlated with one another.

Go to Analyze → Correlate and run a bivariate correlation matrix with the following variables: cap_rate, units, effage, deprec, und_park, elevator, lot_area, area_unt, exp_egi, nbhd1, nbhd3, nbhd4, conditn1, conditn3, story2, story4, and sml_unit.

Correlations								
	cap_rate	units	effage	deprec	und_park	elevator	lot_area	area_unt
cap_rate	1	.548	.744	.109	-.267	-.100	-.101	.377
units	.548	1	.290	-.250	.135	.226	-.169	.265
effage	.744	.290	1	.500	-.545	-.485	.254	.451
deprec	.109	-.250	.500	1	-.553	-.476	.522	.308
und_park	-.267	.135	-.545	-.553	1	.629	-.501	-.118
elevator	-.100	.226	-.485	-.476	.629	1	-.379	-.155
lot_area	-.101	-.169	.254	.522	-.501	-.379	1	-.120
area_unt	.377	.265	.451	.308	-.118	-.155	-.120	1
exp_egi	.051	.339	-.136	-.433	.130	.185	-.216	-.090
nbhd1	-.526	-.238	-.383	-.083	.235	.101	-.140	-.117
nbhd3	-.085	.060	-.443	-.410	.537	.375	-.265	-.172
nbhd4	.304	-.224	.375	.395	-.482	-.313	.462	.138
conditn1	.620	.425	.536	.367	-.247	-.033	.149	.524
conditn3	-.459	-.200	-.508	-.334	.331	.211	-.127	-.054
story2	-.402	-.402	-.123	.280	-.323	-.313	.383	-.266
story4	-.257	-.196	-.131	.164	.036	-.036	-.079	.067
sml_unit	-.714	-.703	-.325	.135	-.067	-.336	.234	-.276

Correlations									
	exp_egi	nbhd1	nbhd3	nbhd4	conditn1	conditn3	story2	story4	sml_unit
cap_rate	.051	-.526	-.085	.304	.620	-.459	-.402	-.257	-.714
units	.339	-.238	.060	-.224	.425	-.200	-.402	-.196	-.703
effage	-.136	-.383	-.443	.375	.536	-.508	-.123	-.131	-.325
deprec	-.433	-.083	-.410	.395	.367	-.334	.280	.164	.135
und_park	.130	.235	.537	-.482	-.247	.331	-.323	.036	-.067
elevator	.185	.101	.375	-.313	-.033	.211	-.313	-.036	-.336
lot_area	-.216	-.140	-.265	.462	.149	-.127	.383	-.079	.234
area_unt	-.090	-.117	-.172	.138	.524	-.054	-.266	.067	-.276
exp_egi	1	-.127	.156	.019	.063	.290	-.266	-.078	-.075
nbhd1	-.127	1	-.302	-.276	-.208	.224	.118	.200	.333
nbhd3	.156	-.302	1	-.333	-.146	.337	.024	-.161	.000
nbhd4	.019	-.276	-.333	1	.234	-.247	.068	-.147	-.079
conditn1	.063	-.208	-.146	.234	1	-.340	-.256	-.203	-.484
conditn3	.290	.224	.337	-.247	-.340	1	.176	.239	.268
story2	-.266	.118	.024	.068	-.256	.176	1	-.147	.394
story4	-.078	.200	-.161	-.147	-.203	.239	-.147	1	.267
sml_unit	-.075	.333	.000	-.079	-.484	.268	.394	.267	1

The correlation between units and cap_rate indicates that capitalization rate increases as the number of units increases. However, there is an even stronger correlation between sml_unit and cap_rate. This indicates that the assumption made earlier concerning buyer motivation may be correct: that there are two types of investors for this type of property: those who invest in buildings with a large number of units and have a short-term outlook, and those who invest in buildings with a smaller number of units and have a longer-term outlook. Also, the negative correlation indicates that there may be a greater demand for small unit apartment buildings, which forces the cap_rate down.

Effage and cap_rate are positively correlated; as the effective age increases, the capitalization rate increases (higher risk). The poor condition binary variable (conditn1) and cap_rate are

positively correlated (0.620), while the above average condition binary variable (conditn3) and cap_rate show a negative correlation (−0.459). That is, as condition improves, the capitalization rate decreases (lower risk).

Among the independent variables, there are some relationships that may cause problems of collinearity in the model calibration.

Effage, the conditn binary variables, and deprec are related. Effage is also related to und_park, elevator, area_unt, and nbhd3. The conditn binary variables are related to most of the same variables as effage but to a lesser degree. Sml_unit is related to units, as would be expected. Care will have to be taken in the variable selection process to avoid distorting the process by including related variables in the same regression run.

Let's review. We want to include the neighbourhood and story binary variables in our initial models. We also want to account for the size of the building and we have two choices, units and the sml_unit binary; but since they are correlated they cannot be in the same model. The effage variable and the condition binary variables account for the same thing, so they should not be placed in the same model either. The lot_area and area_unt variables have some relationship to the cap_rate and provide measures that account for the size of the units and the size of the lot respectively, so we will include both of them. And since, in general, we want more variables rather than fewer in our initial models, we will include the underground parking and elevator variables to round them out.

Thus, the final list of variables for our initial models will be:

Either: units or sml_unit
 Either: effage or conditn binary variables
 Common: und_park
 elevator
 lot_area
 area_unt
 nbhd binary variables
 story binary variables

Variable Selection

The first step in the variable selection process is to use regression analysis to evaluate the initial models. The process followed here will be similar to that used earlier in the course. Each of the groups of candidate variables will be selected as independent variables for the “Enter” regression process, to determine which group produces the best or greatest valued R^2 . As there are two alternative variables for units and two alternatives to represent age or condition, there will have to be four separate groups of independent variables (one group for each combination of alternatives). The dependent variable cap_rate.

The nine variables that will be common to each group will be:

und_park
 elevator
 lot_area
 area_unt
 nbhd binary variables (3)
 story binary variables (2)

The following unique combinations will complete the four groups:

Group 1 units and effage (11 variables)
 Group 2 units and the conditn binary variables (12 variables)
 Group 3 sml_unit and effage (11 variables)
 Group 4 sml_unit and the conditn binary variables (12 variables)

So, the first group will include the following variables:

units	effage	und_park
elevator	lot_area	area_unt
nbhd1	nbhd3	nbhd4
story2	story4	

Select Analyze → Regression → Linear and enter the above variables as Independent variables and cap_rate as the Dependent variable. Then click the Statistics button, and select Estimates, Model Fit, and Collinearity Diagnostics. Then select Continue → Paste. This will place the SPSS statements to run regression into a syntax file (or append them to the bottom of a syntax file if there is one already open). Since most of the variables will be used for all four runs we will be able to copy and paste the regression syntax statements, make a few minor modifications to the variable list and thus speed up the process of running the four initial models. Select the regression statements in the syntax window and run it.



Helpful Hint

This syntax file procedure for running regressions can save you time when you need to run the same or similar regressions for a number of different models, since an SPSS syntax file can be saved and then reopened with a different database. This will be helpful when you are working on the Case Study exam. However, if you prefer instead to use the “point and click” features in the Linear Regression window, you will obtain exactly the same output.

For the second test, if you are using the syntax method, copy and paste a second copy of the regression statements after the first set of statements, locate effage in the variable list near the end of the statements and replace it with the two conditn binary variables. Select the new set of statements and re-run the regression.

The next two tests will use sml_unit instead of units; for the final test the two conditn binaries should be replaced with effage re-entered.

A summary table of results from the four initial models is below:

Group	Variables (plus common)	R	R ²	Adjusted R ²	SEE	F	Sig.	Maximum VIF
1	UNITS/EFFAGE	0.934	0.873	0.795	0.00553	11.207	0.000	3.210
2	UNITS/CONDITN binaries	0.883	0.780	0.624	0.00749	5.016	0.001	3.476
3	SML_UNIT/EFFAGE	0.954	0.910	0.855	0.00465	16.524	0.000	3.208
4	SML_UNIT/CONDITN binaries	0.905	0.819	0.691	0.00679	6.401	0.000	3.469

Of the four alternatives, Group 3, with effage and sml_unit produces the greatest R² statistic as well as the least valued SEE. As well, Group 2 and Group 4 suffer from some minor multicollinearity – both have a maximum VIF statistic that is slightly greater than 3.33. We will continue with Group 3, the main tables from its regression run are shown below:

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.954 ^a	.910	.855	.00465

a. Predictors: (Constant), story4, elevator, area_unt, nbhd1, nbhd4, story2, lot_area, sml_unit, nbhd3, effage, und_park

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.004	11	.000	16.524	.000 ^b
	Residual	.000	18	.000		
	Total	.004	29			

a. Dependent Variable: cap_rate

b. Predictors: (Constant), story4, elevator, area_unt, nbhd1, nbhd4, story2, lot_area, sml_unit, nbhd3, effage, und_park

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	.035	.011		3.017	.007		
	sml_unit	-.010	.003	-.422	-4.040	.001	.460	2.175
	effage	.001	.000	.641	5.118	.000	.319	3.135
	und_park	-.005	.003	-.214	-1.688	.109	.312	3.208
	elevator	-.001	.003	-.031	-.277	.785	.396	2.527
	lot_area	-.008	.004	-.194	-2.049	.055	.560	1.786
	area_unt	-8.167E-6	.000	-.098	-1.144	.268	.688	1.454
	nbhd1	.001	.003	.045	.403	.692	.401	2.493
	nbhd3	.009	.003	.322	2.595	.018	.326	3.069
	nbhd4	.004	.003	.151	1.661	.114	.608	1.644
	story2	-.006	.003	-.216	-2.251	.037	.546	1.832
	story4	-.001	.004	-.030	-.368	.717	.772	1.296

a. Dependent Variable: cap_rate

With only 30 sales in the database, the number of variables that we include in the final model is very important. A general “rule of thumb” is to have a minimum of five sales in the database for each variable included in the final model. In the present case this is very difficult, as there are three neighbourhood variables alone and two for the number of stories. In this instance, we will relax the “minimum of five” rule somewhat by considering the three neighbourhood binary variables as one variable, and the two story binary variables as one as well, this gives us a count of eight variables in the current model.

The statistics shown above indicate there are no major multicollinearity problems (all VIF statistics are less than 3.333), but several of the variables have low t-statistics (i.e., inside the range of ± 1.6). There are a couple of methods that we could use to reduce the variable count from eight to six but we must be diligent to ensure that all of the included variables have reasonable coefficients and are not in conflict.

One method is to manually remove the variables that have the t-statistics which are closest to zero. Those would be, in order: elevator (−0.277), story4 (−0.368), nbhd1 (0.403), area_unt (−1.144), and nbhd4 (1.661). Removing the story or neighbourhood binary variables would not affect the variable count since story2 and nbhd3 would remain in the model. With that in mind, our best course of action would be to remove the elevator and area_unt variables and re-run the regression leaving sml_unit, effage, lot_area, und_park, the nbhd binaries, and the story binaries for a count of six. This would yield an adjusted R^2 of 0.860 and an SEE of 0.00458.

A second method, and the one we will use here, is to use stepwise regression. We will apply appropriate selection criteria in order to remove some of the less significant variables.

Following the steps recommended above, the Group 3 model is the one with which we will continue, and, therefore, only minor changes are needed to the syntax statements for the stepwise process. Copy and paste the Group 3 regression statements to the bottom of the syntax file, and make the following changes:

Change PIN(.05) to PIN(.10) and POUT(.10) to POUT(.15)

Change /METHOD=ENTER to /METHOD=STEPWISE



Helpful Hint

If you are using the Linear Regression window, under Method select Stepwise instead of Enter, then click on Options, and change Entry to .10 and Removal to .15.

Block and run the new regression statements. Note that only five variables have been selected, and one of the variables not selected, lot_area, has a probability that is right at the threshold of acceptance (a Sig. value of 0.150 – see the final step of the Excluded Variable table).

Increase the PIN (entry requirement) to .15, and increase the POUT (removal requirement) to .20, and re-run the regression. The following is the coefficient section of the final report:

Coefficients ^a								
		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
Model		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.006	.009		.652	.520		
	effage	.001	.000	.744	5.890	.000	1.000	1.000
2	(Constant)	.024	.007		3.534	.001		
	effage	.001	.000	.572	6.320	.000	.894	1.118
	sml_unit	-.013	.002	-.527	-5.824	.000	.894	1.118
3	(Constant)	.015	.007		1.984	.058		
	effage	.001	.000	.679	7.197	.000	.698	1.433
	sml_unit	-.012	.002	-.493	-5.822	.000	.868	1.152
	nbhd3	.006	.002	.216	2.419	.023	.780	1.281
4	(Constant)	.015	.007		2.077	.048		
	effage	.001	.000	.683	7.576	.000	.697	1.434
	sml_unit	-.010	.002	-.431	-4.939	.000	.745	1.343
	nbhd3	.006	.002	.221	2.594	.016	.780	1.283
	story2	-.004	.002	-.154	-1.876	.072	.844	1.185
5	(Constant)	.024	.008		2.901	.008		
	effage	.001	.000	.587	5.927	.000	.521	1.918
	sml_unit	-.011	.002	-.447	-5.369	.000	.737	1.356
	nbhd3	.008	.002	.292	3.285	.003	.647	1.545
	story2	-.006	.002	-.228	-2.628	.015	.678	1.474
	und_park	-.005	.003	-.207	-1.933	.065	.443	2.256
6	(Constant)	.028	.009		3.311	.003		
	effage	.001	.000	.606	6.222	.000	.512	1.954
	sml_unit	-.010	.002	-.425	-5.153	.000	.714	1.400
	nbhd3	.008	.002	.287	3.314	.003	.647	1.546
	story2	-.006	.002	-.198	-2.278	.032	.642	1.557
	und_park	-.006	.003	-.247	-2.288	.032	.416	2.401
	lot_area	-.005	.003	-.127	-1.491	.150	.666	1.502
7	(Constant)	.029	.008		3.567	.002		
	effage	.001	.000	.600	6.400	.000	.511	1.957
	sml_unit	-.010	.002	-.407	-5.094	.000	.702	1.424
	nbhd3	.008	.002	.296	3.544	.002	.644	1.552
	story2	-.005	.002	-.183	-2.180	.040	.635	1.574
	und_park	-.005	.003	-.207	-1.939	.065	.396	2.528
	lot_area	-.007	.004	-.178	-2.040	.054	.588	1.700
	nbhd4	.004	.002	.141	1.705	.102	.655	1.526

a. Dependent Variable: cap_rate

Loosening the selection criteria adds not only lot_area but also nbhd4. The *t*-statistics improve for both these variables so that only nbhd4 is insignificant at the 10% level, and the probability is only slightly greater at 10.2%. These changes have also improved the adjusted R^2 and SEE statistics. Finally, if the nbhd variables are counted as one variable, there are only six variables in the model, which meets the variable count “rule-of-thumb”.

Students should note that our first method of manually removing the two variables that were not nbhd nor story related resulted in a slightly better R^2 , but a slightly worse adjusted R^2 and a slightly greater SEE (0.903, 0.860, and 0.00458 versus 0.901, 0.870, and 0.00441).

Analysis of Results

At this point, we will run a final regression and include only the relevant variables from the final step in the stepwise process. But first, we will make a few additional changes to the syntax file.

As indicated in the introduction to this lesson, the results of this model will be examined against the sales information contained in the database and also against traditional single property methods. To do so we will need to store the estimated capitalization rates in the database; when we re-run the model, we will save the (unstandardized) predicted values.

Copy the final set of regression statements and paste them at the bottom of syntax file and make the following changes:

on the /METHOD line: change STEPWISE to ENTER
 remove elevator, area_unit, nbhd1, and story4
 remove the period (.) from end of the line
 add a new bottom line: /SAVE PRED.



Helpful Hint

To perform these steps in the Linear Regression window, remove the four noted variables, then, click Save, and select Unstandardized under Predicted Values.

Run the new set of regression statements. The Coefficients table should look something like this; the number of decimal places has been increased from 3 to 5 for the coefficients (the numbers in the column labelled B):

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
	B	Std. Error	Beta				Tolerance	VIF
1	(Constant)	.02947	.008		3.567	.002		
	sml_unit	-.00978	.002	-.407	-5.094	.000	.702	1.424
	effage	.00108	.000	.600	6.400	.000	.511	1.957
	und_park	-.00500	.003	-.207	-1.939	.065	.396	2.528
	lot_area	-.00719	.004	-.178	-2.040	.054	.588	1.700
	nbhd3	.00803	.002	.296	3.544	.002	.644	1.552
	nbhd4	.00401	.002	.141	1.705	.102	.655	1.526
	story2	-.00520	.002	-.183	-2.180	.040	.635	1.574

a. Dependent Variable: cap_rate

When finished, access the Variable View worksheet and rename the PRE_1 variable to estcrate (estimated cap rate) by typing the new name on top of the old one. Then run the following transformations:

COMPUTE cr_ratio=estcrate/cap_rate. (ratio of estimated cap rate to actual cap rate)
 COMPUTE estprice=noi/estcrate. (price estimate using estimated cap rate)
 COMPUTE epratio=estprice/price. (ratio of estimated price to actual price)

Use Analyze → Descriptive Statistics → Frequencies to produce the following statistics for cr_ratio and epratio. Also select Graphs → Legacy Dialogs → Histograms in order to view if there are any outliers.

Following is a summary of the results:

Statistics			
		cr_ratio	epratio
Mean		1.0039	1.0008
Median		1.0075	.9926
Std. Deviation		.06829	.07205
Percentiles	10	.9204	.9124
	25	.9633	.9476
	75	1.0553	1.0381
	90	1.0960	1.0865

The median and the mean are very close to one for both of these ratios. In addition, the standard deviation for each is small relative to their mean; that is, the COVs are small (less than 10%) – 6.8% and 7.2% respectively. These results show that this method of estimating cap_rate produces a very good result, not only for the capitalization rate itself, but also for using this capitalization rate to estimate the sale price.

At this point further testing can be carried out to find out how good the model really is. For example, is the cr_ratio equitable across the neighbourhoods, or storey counts, or the condition values? Tools such as K-W and M-W tests, as well as ratio statistics with a grouping variable should be run to establish the quality of the model and to discover if any adjustments are needed. These testing procedures have been demonstrated in earlier lessons and as such, we'll leave it to the reader to test the cr_ratio and epratio.

Comparison with Single Property Techniques

In order to test the results of the regression model against the results from traditional single property techniques, capitalization rates will first have to be developed by applying these single property techniques. The sale properties must be separated into subgroups of comparable properties that represent the various capitalization rates that are found in this real estate market. For example, if purchasers of income properties apply a distinctly different capitalization rate to properties with two stories compared to properties with four stories, this difference would have to be recognized by the appraiser when calculating the capitalization rate for a given subject property. One stratification developed earlier was buildings with either small or large numbers of units. It would seem to make sense that if an appraiser was valuing a 10-unit apartment building, he or she would not look for comparables among 100-unit buildings. To develop accurate capitalization rates, the appraiser must stratify the comparables according to market preferences and appraisal theory. We will examine the capitalization rates of the sale properties in the Apartment database separated first by the number of units in the building (the sml_unit binary variable developed earlier) and then by other secondary stratifications that could be used by an appraiser in this market.

Generally, single property techniques require a minimum of three to five comparable properties to adequately estimate a market capitalization rate. We will test story, und_park, elevator, conditn and nbhd as secondary stratifications of the sml_unit variable, but these subgroups will only be used if each contains at least three sales. As conditn is highly correlated to effage, conditn will be used instead of creating age groupings, even though the regression analysis suggests that effage is a better variable in explaining variations in cap_rate. Students may wish to create three or four age groups, and compare the results to those using conditn, but this will not be illustrated in this lesson.

Set a filter to select properties for which sml_unit =1. Then go to Data → Split File, check “Organize output by groups”, and enter story as the Groups Based On variable. This splits the database into three groups: one for each story height. Now, using Analyze → Descriptive Statistics

→ Frequencies run a frequency distribution on cap_rate and record the median values for each of the three groups.



Helpful Hint

To clear out an old setting in Frequencies, click the Reset button. The only statistic required is the median. Uncheck “Display frequency tables” to streamline the output.

Return to Split File and repeat the analysis for each of the other four grouping variables (und_park, elevator, conditn, and nbhd), recording the median value and number of observations in each subgroup.

When finished, change the filter to select cases for which sml_unit = 0 and repeat the process. The following table shows the results.

		18 units or less (SML_UNIT)			
		1		0	
		Median	Observations	Median	Observations
No Break		0.0492	15	0.0661	15
Story	2	0.0482	6	0.0493	1
	3	0.0504	7	0.0669	14
	4	0.0463	2	-	0
Neighbourhood	1	0.0457	5	0.0496	1
	2	0.0548	3	0.0669	6
	3	0.0475	4	0.0624	4
	4	0.0522	3	0.0782	4
Underground Parking	0	0.0522	9	0.0735	8
	1	0.0468	6	0.0609	7
Elevator	0	0.0522	9	0.0751	4
	1	0.0454	6	0.0609	11
Condition	1	0.0489	2	0.0745	9
	2	0.0518	9	0.0595	5
	3	0.0454	4	0.0496	1

These results describe the capitalization rates found using traditional single property appraisal techniques under each of the secondary stratifications for both small and large apartment buildings. For example, the overall capitalization rate for small buildings was 4.9% and 6.6% for large buildings. The capitalization rate for small buildings that had underground parking was 4.7%, and 5.2% for small buildings that did not have underground parking. Similarly, large buildings with and without underground parking had capitalization rates of 6.1% and 7.4%, respectively. The stratifications that are viewed as significant will be used to determine the capitalization rates for single property appraisal.

The table shows that several subgroups have less than three observations in each group, which eliminates that subgroup from the process, based on the established minimum of three sales. For buildings with a small number of units, the nbhd, und_park, and elevator groups all have an adequate number of sales, while the buildings with a large number of units only have sufficient sales in the und_park and elevator groups. This suggests that the capitalization rate estimates for buildings with small or large numbers of units could be further separated into subgroups based on either und_park, elevator, or, possibly, nbhd. Since there are only 30 comparable sales in total, it is unlikely that there will be enough comparables to create further subgroups once this secondary stratification is chosen. Appraisal judgement suggests that among these potential variables neighbourhood is the most important factor for appraisals when determining capital-

ization rates. We will use the nbhd variable to stratify the capitalization rates for building with small and large numbers of units. Since there are not enough observations of buildings with a large number of units in neighbourhood 1, we cannot rely on the median cap_rate calculated. Instead, a median value must be estimated for this subgroup. In most cases in this database, the median cap_rate for buildings with a large number of units is approximately 1.5% greater than the median for buildings with a small number of units. This is true for the overall database as well as the individual subgroups. Therefore, the capitalization rate estimate for the building in neighbourhood 1 with a large number of units can be set about 1.5% greater than the rate for buildings in neighbourhood 1 with a small number of units. A such, a capitalization rate of 6% will be used for this subgroup.

There are two methods to enter these rates in the database. As there are only 30 sales, we could create a report listing the sale number, nbhd, and sml_unit, then write in the appropriate cap_rate on the report, and manually enter them into a new variable. Alternatively, we can create a transformation to do this. The transformation method will be shown here. The variables which will be created are shown in the following transformations. Ensure you remove the filter and split file setting before proceeding.

```
COMPUTE nbhdcr_s=nbhd.
RECODE nbhdcr_s(1=.0457)(2=.0548)(3=.0475)(4=.0522).
COMPUTE nbhdcr_l=nbhd.
RECODE nbhdcr_l(1=.0600)(2=.0669)(3=.0624)(4=.0782).
COMPUTE scaprate=nbhdcr_s.
IF (sml_unit=0)scaprate=nbhdcr_l.
COMPUTE sesprice=noi/scaprate.
COMPUTE sesratio=sesprice/price.
```

Nbhdcr_s and nbhdcr_l are variables that describe the capitalization rates by neighbourhood for buildings with small and large numbers of units. Scaprate combines the previous two variables into one variable with appropriate values for each type of building. The final two variables create a price estimate and ratio of estimated price to actual price. The following table summarizes the ratio statistics for sesratio, first for all sales, and then filtering out the one low outlier found.

	SESRATIO Statistics	
	With Outlier	Without Outlier
Mean	0.987	1.000
Median	1.000	1.001
Standard Deviation	11.13%	9.00%
25% to 75%	0.945 to 1.038	0.955 to 1.041
10% to 90%	0.833 to 1.126	0.835 to 1.130

This method produces a reasonable result and both the mean and median values are very close to 1. The standard deviation is within acceptable limits, but it is considerably larger than that produced by the regression model. This is true even when the one outlier is removed. The COVs are 11.3% and 9.0% respectively.

It is possible that the outlier could have been caused by our estimate of the capitalization rate for the one large unit in neighbourhood 1 being too large. An alternative to the method used would be to not split neighbourhood 1 into large and small number of units. This would result in a median capitalization rate for neighbourhood 1 that is only slightly greater than the one used for the small number of units, and much closer to the actual capitalization rate for the one large unit. If the procedures above were run again for neighbourhood 1 without splitting this variable into groups by number of units, the result would not improve significantly compared to the above results with the outlier. The outlier would remain, and the standard deviation would only be reduced to 10.9%. This implies that our capitalization rate estimate for the neighbourhood 1 large unit subgroup did not have a significant effect on the capitalization rates estimated.

The results above show that using multiple regression analysis to estimate the capitalization rate in a small sample will produce a considerably improved estimate of value over using the single property method of capitalization rate selection. This method results in less than two-thirds of the dispersion that is found using the best estimate from the single property method when all of the sales are included. When the outlier is removed, the dispersion using regression analysis is still 20% lower.

CASE STUDY

We'll finish off this lesson with a Case Study drawn from a western Canadian city. The assessment department uses multiple regression for their income modeling. While the data is protected for privacy reasons, the methodology used for the valuation of apartment buildings will be briefly discussed here.

The Model

The approach is made up of five sub-models, one each for rent, vacancy and bad debts, other income, operating expenses, and overall capitalization rate (OCR). The rent model uses multiplicative regression analysis while the rest use additive regression analysis.

Mail-out questionnaires are used to gather rents, vacancy rates, and operating income. Additional information can be drawn from Canada Mortgage and Housing rental market reports as well as revisions or appeals. During the study period more than 250 apartment buildings were sold; sales information was collected from the provincial Land Titles Office.

The market value is the net annual operating income divided by the overall capitalization rate. The net annual operating income is defined as follows:

Potential Annual Rent – Vacancy and Bad Debt Loss + Other Income – Operating Expenses.

The first step is to generate a rent model with the potential gross income per unit as the dependent variable. The potential gross income includes the potential rent plus income from parking and laundry facilities. Base rents are generated for four suite sizes: bachelor, 1-bedroom, 2-bedroom, and 3- or 4-bedroom. Multiplicative adjustments are applied to the base rents according to various factors including:

- age grouping;
- size of building in groups based on number of units;
- number of storeys (3 or fewer, 4 to 6, 7 to 9, and 10 or more);
- a subjective measure based on the building's tenant appeal (poor to excellent); and
- neighbourhood

At this point, it should be pointed out that there is some danger of multicollinearity between the appeal of the building and the neighbourhood: an apartment building in a more upscale neighbourhood may automatically have a higher appeal over a similar building in an older neighbourhood.

The vacancy and bad debt model generated different percentages based on neighbourhood. The other income model had base rates for different combinations of the availability of parking and laundry facilities with additive adjustments for age group, storey group, and neighbourhood. Similarly, the fourth sub-model, a base percentage rate for operating expenses, was developed with additive adjustments for the same categories (age group, storey group, and neighbourhood).

Itemizing the models gives the following formula for the Net Annual Operating Income:

Potential Annual Rent

$$\sum_{n=1}^4 (\text{suitetype}_n \times \text{adjustedrent}_n)$$

MINUS Vacancy and Bad Debt Loss

Potential Annual Rent × neighbourhood rate

PLUS Other Income (parking and laundry)

Base Rate + age group, storey group, and neighbourhood adjustments

MINUS Operating Expenses (a percentage of the Potential Annual Rent)

Potential Annual Rent × (Base Rate + age group, storey group, and neighbourhood adjustments)

Finally, the overall capitalization rate model was developed based on the adjusted sale prices. It yielded a base rate of 9.0% with a 2.5% adjustment for the oldest buildings (11.5%) and an additional 1.0% adjustment for three of the neighbourhoods. Thus, the range in the overall capitalization rate was 9.0% to 12.5%.

The predicted value for the properties in the study was calculated... $\text{NOI} \div \text{OCR}$...and yielded the following ratio statistics (ASR: predicted to actual sale price).

Mean	0.913
Median	0.902
Minimum	0.537
Maximum	1.598
Coefficient of Dispersion	0.137

Observations

Generally, the results look satisfactory; the COD is very good for income generating property such as apartment buildings. However, if the target median ASR is 1.00, an overall adjustment of roughly 10% is needed. Such an adjustment would change the minimum/maximum to 0.584/1.758. It is likely that finer stratification, such as by neighbourhood or age grouping, would result in more targeted adjustments which would narrow the overall range of the ASR.

One thing to note here for students is the potential for multicollinearity in the overall model. Property characteristics such as neighbourhood, age grouping, and storey grouping are accounted for in different ways in various sub-models. Care must be taken to ensure that the sub-models are well isolated from each other. Solid model specification and calibration as well as a thorough testing regimen will reduce any exposure to multicollinearity that could result within the model.

SUMMARY

In this lesson you have stated a general income model and proceeded to examine the variables in the database using descriptive statistics and graphs. A number of transformations were then carried out to get some variables into the right form for additive multiple regression, most notably the creation of binary variables for neighbourhood, storey count, and condition.

Four initial models were run and the best model was selected based on the statistical results (R^2 and SEE). The chosen model was then calibrated and an overall test was carried out showing that the final model was acceptable.

The model also fared quite well against our results from our use of a traditional single property appraisal method.

The lesson finished off with an examination of an actual application of regression analysis applied to apartment building in a western Canadian city.

At this point in the course, you should be comfortable with multiple regression analysis and its application to a broad range of property types. Additionally, you should be aware of some of the pitfalls that can seriously undermine this kind of analysis, and how to avoid them.

REVIEW AND DISCUSSION QUESTIONS

The following questions are for review and discussion purposes only. They are not to be submitted as part of the assignment for this lesson. Answers to select questions can be found on the Course Resources webpage. However, in order to develop your understanding, you are encouraged to attempt answering the questions before accessing the answers. You may also wish to discuss these questions with your fellow students on the course discussion forum – through discussing these questions, you will better understand the course materials and be better prepared for the final examination.



Note: Use the Apartment.sav file to answer the following questions.

1. Calculate the coefficient of variation (standard deviation $\div$ mean) for the gim, sp_unit, and sp_sqft variables to determine what is the best market value indicator for this database.
2. Create an additive multiple regression model to estimate the value of the unit of measure you selected in Question 1. Describe the steps you followed to create your final model and show all pertinent output. In addition, you are also expected to briefly test and evaluate the model, and state your conclusions as to the quality of the model.

Note: Considerable caution should be exercised in the variable selection for the model in this question. It is extremely important that the variables to be used in the model are independent of each other and of the dependent variable selected. For example, if the dependent variable chosen is sp_sqft, the sale price and square foot variables would be completely correlated with sp_sqft and could not be used in a regression model with sp_sqft as the dependent variable. In fact, any variable that was created using sale price is not a suitable independent variable for use in this model. Since this model is simply a preliminary step in developing value estimates for income-producing properties, using a variable such as price, or any variable which requires price for its development, will render the model useless for valuation of properties that have not sold.

3. The third step of the nine model building steps: Examine the potential independent variables for relationships with each other and with the dependent variable using graphic analysis, crosstabs, and correlation analysis. Briefly describe how you would examine the variables in your database and explain what in particular you are looking for in term of suitable variables for your income model. What makes a good variable for inclusion in the final model?
4. The main components that make up a discount or capitalization rate are: risk, liquidity, and investment management. Briefly describe these three components.
5. Gross income multipliers (gim) and capitalization rates (cap rate) can be calculated using multiple regression analysis (MRA) or a spreadsheet analysis. Provide some advantages for the MRA approach.

ASSIGNMENT 7

Lesson 7: Income-Based Modeling

Marks: 1 mark per question

1. The income approach to valuing real property commonly uses which of the following formulas?
 - (1) Market Value = Net Operating Income/Overall Capitalization Rate
 - (2) Market Value = Gross Operating Income/Overall Capitalization Rate
 - (3) Gross Operating Income = Net Operating Income/Overall Capitalization Rate
 - (4) Gross Operating Income = Overall Capitalization Rate/Tax Assessment Value

2. Which of the following is a reason to remove outliers from a linear regression model?
 - (1) The low outliers will consistently overcorrect for the high outliers, leading to a biased regression model.
 - (2) Outliers will likely have a significant effect on the outcome of the regression model, and may lessen its predictive power.
 - (3) Outliers contribute to the statistical problem of “overfitting”.
 - (4) None of the above

3. In the model developed in Lesson 7, we found that purchasers of large buildings require a higher cap rate than do purchasers of smaller buildings. Which of the following statements most accurately explains why?
 - (1) Large buildings are riskless investments.
 - (2) Small buildings are often extremely risky investments.
 - (3) Small buildings typically require a greater number of lifetime repairs.
 - (4) There may be lower demand for large buildings, which results in higher NOI-to-price ratios and, thus, higher cap rates.

4. Which of the following statements about correlation coefficients is TRUE?
 - (1) They can never be negative.
 - (2) They are equivalent to R^2 .
 - (3) They are always positive.
 - (4) They can be positive or negative.

5. Which of the following statements about boxplots is TRUE?
 - (1) Boxplots provide the data transformations required for most regression models.
 - (2) Boxplots are rarely useful because they show means only, and cannot represent median values.
 - (3) Boxplots require linear regression model output before they can be created.
 - (4) None of the above statements about boxplots is true. They are all false.

6. Look at the coefficients table from the final multiple regression model report in this lesson. Which independent variables increase the capitalization rate?
 - (1) lot_area, nbhd3, nbhd4, effage and sml_unit
 - (2) nbhd3, nbhd4 and effage
 - (3) story2, lot_area, und_park and sml_unit
 - (4) sml_unit, nbhd3, nbhd4 and effage

Assignment continues on the following page

ASSIGNMENT 7, continued

7. Look at the coefficients table from the final multiple regression model report in this lesson. Which independent variable is the most significant?
 - (1) lot_area
 - (2) nbhd3
 - (3) sml_unit
 - (4) effage
8. Why do investors choose income producing real estate from a wide array of investment opportunities?
 - (1) Cash flow
 - (2) Property value appreciation (inflation)
 - (3) Income tax savings (deferred taxes)
 - (4) All of the above
9. Which of the following best describes the reason for re-setting the values of PIN and POUT in multiple regression analysis?
 - (1) To remove multicollinearity from the equation.
 - (2) To apply appraisal judgment on the variable selection process.
 - (3) To relax or tighten the significant criterion to either remove or include variables in a regression.
 - (4) To decrease the R^2 value.
10. A capitalization rate may be influenced by the level of risk involved in an investment property. If the level of risk rises, the capitalization rate will generally tend to:
 - (1) decrease.
 - (2) increase.
 - (3) stay the same.
 - (4) become useless.



Planning Ahead

Project 2 is due in one week. You should be well into the project by now. If you are having any difficulties with Project 2, please contact the course tutor (sooner rather than later).

End of assignment

